

SCHEME OF VALUATION

Course Title: Engineering Mathematics-II

Examination: MID-II

Academic Year: 2025-26

Max. Marks: 30

PART – A ($3 \times 2 = 6$ Marks)

1(a) Find the orthogonal trajectories of the given family of parabolas.

- Formation of differential equation – 1 Mark
- Orthogonal trajectory obtained correctly – 1 Mark

Total: 2 Marks

1(b) Solve the given differential equation.

- Correct method/formulation – 1 Mark
- Correct solution – 1 Mark

Total: 2 Marks

1(c) Find the Laplace transform of the given function.

- Application of Laplace transform formula – 1 Mark
- Correct result – 1 Mark

Total: 2 Marks

PART – B ($2 \times 12 = 24$ Marks)

2(a) Solve the given differential equation.

- Identification of method – 1 Mark
- Correct procedure – 3 Marks
- Simplification and final answer – 2 Marks

Total: 6 Marks

2(b) Solve the given differential equation.

- Correct formulation – 2 Marks
- Solution process – 3 Marks
- Final answer – 1 Mark

Total: 6 Marks

OR

3(a) Construct the differential equation for the orthogonal trajectories of the given family of curves.

- Elimination of parameter – 2 Marks
- Formation of differential equation – 2 Marks

- Orthogonal trajectory obtained – 2 Marks

Total: 6 Marks

3(b) Newton's Law of Cooling Problem.

- Formation of differential equation – 2 Marks
- Determination of constant using given data – 2 Marks
- Temperature after 40 minutes – 2 Marks

Total: 6 Marks

4(a) Solve the given differential equation.

- Complementary function – 2 Marks
- Particular integral – 2 Marks
- Complete solution – 2 Marks

Total: 6 Marks

4(b) Solve by Variation of Parameters Method.

- Complementary function – 2 Marks
- Determination of parameters – 2 Marks
- Final solution – 2 Marks

Total: 6 Marks

OR

5(a) Using Convolution Theorem, find the inverse Laplace transform.

- Laplace transform decomposition – 2 Marks
- Application of convolution theorem – 2 Marks
- Final answer – 2 Marks

Total: 6 Marks

5(b) Solve using Linear Transformation.

- Transformation of variables – 2 Marks
- Reduction to standard form – 2 Marks
- Final solution – 2 Marks

Total: 6 Marks

Note: Award proportionate marks for correct procedure, intermediate steps, and final answers. Minor arithmetic errors may be ignored if the method is correct.